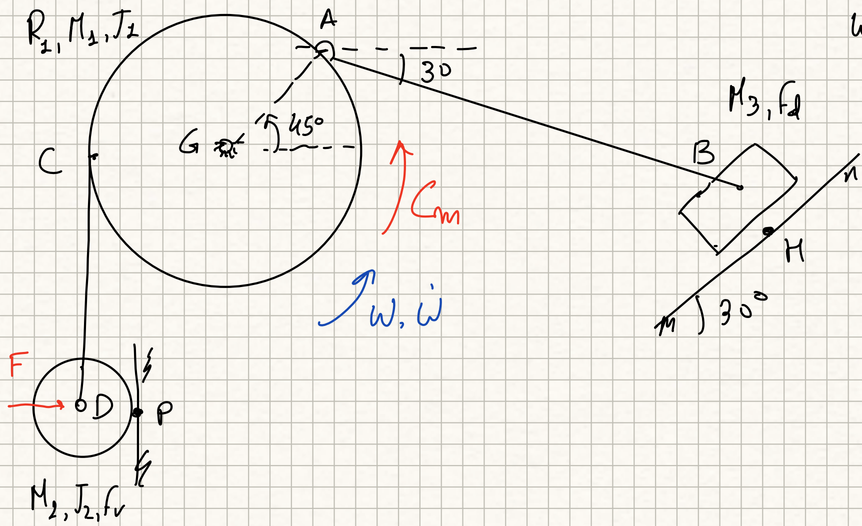


$\omega = 1$ rad/s, $\dot{\omega} = 1$ rad/s^2 $AB = 3$ m

$R_1 = 1$ m , $J_1 = 0.6$ kgm^2

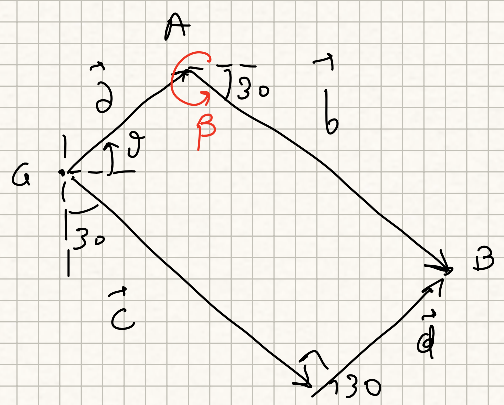
$M_2 = 2$ kg $J_2 = 0.2$ kgm^2 $R_2 = 0.5$ m

$M_3 = 4$ kg $f_d = 0.7$, $f_v = 0{,}05$

$F = 30$ N

$gdl = 3m = 12$

$gdv = 2_C + 2_A + 2_D + 2_P + 2_H + 1_{FUNE}$
$= 11$



const	a	b	c	γ	δ
var	α	β	d		

Posizioni: $a = 1$ m , $b = 3$ m , $c = ?$, $d = ?$

$\alpha = 45°$ $\beta = 330$, $\gamma = 300°$, $\delta = 30°$

$$\begin{cases} a\cos\alpha + b\cos\beta = c\cos\gamma + d\cos\delta \\ a\sin\alpha + b\sin\beta = c\sin\gamma + d\sin\delta \end{cases} \rightarrow$$

$$\begin{bmatrix} \cos\gamma & \cos\delta \\ \sin\gamma & \sin\delta \end{bmatrix} \begin{Bmatrix} c \\ d \end{Bmatrix} = \begin{Bmatrix} a\cos\alpha + b\cos\beta \\ a\sin\alpha + b\sin\beta \end{Bmatrix}$$

$$\underbrace{\begin{bmatrix} 0.5 & \sqrt{3}/2 \\ -\sqrt{3}/2 & 0.5 \end{bmatrix}}_{\det(A) = 1} \begin{Bmatrix} c \\ d \end{Bmatrix} = \begin{Bmatrix} 3.31 \\ -0.79 \end{Bmatrix}$$

$$c = \begin{vmatrix} 3.31 & \sqrt{3}/2 \\ -0.79 & 0.5 \end{vmatrix} = 2.34 \text{ m} \qquad d = \begin{vmatrix} 0.5 & 3.31 \\ -\sqrt{3}/2 & -0.79 \end{vmatrix} = 2.47 \text{ m}$$

Velocità (α, β, d variabili)

$$\begin{cases} -a\dot{\alpha}\sin\alpha - b\dot{\beta}\sin\beta = \dot{d}\cos\delta \\ a\dot{\alpha}\cos\alpha + b\dot{\beta}\cos\beta = \dot{d}\sin\delta \end{cases} \Rightarrow \begin{bmatrix} b\sin\beta & \cos\delta \\ -b\cos\beta & \sin\delta \end{bmatrix} \begin{Bmatrix} \dot{\beta} \\ \dot{d} \end{Bmatrix} = \begin{Bmatrix} -a\dot{\alpha}\sin\alpha \\ a\dot{\alpha}\cos\alpha \end{Bmatrix}$$

$$\underbrace{\begin{bmatrix} -1.5 & \sqrt{3}/2 \\ -2.6 & 0.5 \end{bmatrix}}_{} \begin{Bmatrix} \dot{\beta} \\ \dot{d} \end{Bmatrix} = \begin{Bmatrix} -\sqrt{2}/2 \\ \sqrt{2}/2 \end{Bmatrix} \qquad \dot{\beta} = \frac{\begin{vmatrix} -\sqrt{2}/2 & \sqrt{3}/2 \\ \sqrt{2}/2 & 0.5 \end{vmatrix}}{\det(A)} = -0.64 \text{ rad/s}$$

$\det(A) = 1.5$

$$\dot{d} = \frac{\begin{vmatrix} -1.5 & -\sqrt{2}/2 \\ -2.6 & \sqrt{2}/2 \end{vmatrix}}{\det(A)} = -1.93 \text{ m/s}$$

Accelerazioni (α, β, d)

$$\begin{cases} -a\ddot{\alpha}\sin\alpha - a\dot{\alpha}^2\cos\alpha - b\ddot{\beta}\sin\beta - b\dot{\beta}^2\cos\beta = \ddot{d}\cos\delta \\ a\ddot{\alpha}\cos\alpha - a\dot{\alpha}^2\sin\alpha + b\ddot{\beta}\cos\beta - b\dot{\beta}^2\sin\beta = \ddot{d}\sin\delta \end{cases}$$

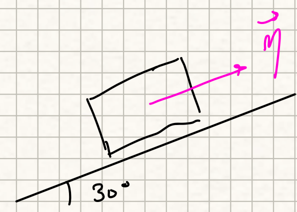
$$\begin{bmatrix} b\sin\beta & \cos\delta \\ -b\cos\beta & \sin\delta \end{bmatrix} \begin{Bmatrix} \ddot{\beta} \\ \ddot{d} \end{Bmatrix} = \begin{Bmatrix} -a\ddot{\alpha}\sin\alpha - a\dot{\alpha}^2\cos\alpha - b\dot{\beta}^2\cos\beta \\ a\ddot{\alpha}\cos\alpha - a\dot{\alpha}^2\sin\alpha - b\dot{\beta}^2\sin\beta \end{Bmatrix}$$

$$\underbrace{\begin{bmatrix} -1.5 & \sqrt{3}/2 \\ -2.6 & 0.5 \end{bmatrix}}_{} \begin{Bmatrix} \ddot{\beta} \\ \ddot{d} \end{Bmatrix} = \begin{Bmatrix} -a\ddot{\alpha}\sin\alpha - a\dot{\alpha}^2\cos\alpha - b\dot{\beta}^2\cos\beta \\ a\ddot{\alpha}\cos\alpha - a\dot{\alpha}^2\sin\alpha - b\dot{\beta}^2\sin\beta \end{Bmatrix}$$

$\det(A) = 1.5$

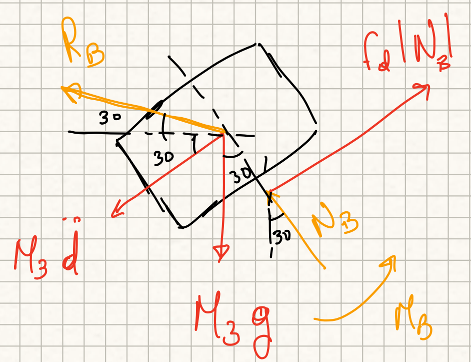
$$\begin{bmatrix} -1.5 & \sqrt{3}/2 \\ -2.6 & 0.5 \end{bmatrix} \begin{Bmatrix} \ddot{\beta} \\ \ddot{d} \end{Bmatrix} = \begin{Bmatrix} -2.48 \\ 0.61 \end{Bmatrix}$$

$$\ddot{\beta} = \frac{\begin{vmatrix} -2.48 & \sqrt{3}/2 \\ 0.61 & 0.5 \end{vmatrix}}{\det(A)} = -1.18 \ rad/s^2 \qquad \ddot{d} = \frac{\begin{vmatrix} -1.5 & -2.48 \\ -2.6 & 0.61 \end{vmatrix}}{\det(A)} = -4.91 \ m/s^2$$



$$\vec{v}_B = \dot{d}\,\vec{\eta} = (-1.93\,\vec{\eta}) \ m/s$$
$$= \dot{d}\left(\cos 30\,\vec{i} + \sin 30\,\vec{j}\right) = \left(-1.67\,\vec{i} - 0.965\,\vec{j}\right) \ m/s$$

$$\vec{a}_B = \ddot{d}\,\vec{\eta} = (-4.91\,\vec{\eta}) \ m/s^2$$
$$= \ddot{d}\left(\cos 30\,\vec{i} + \sin 30\,\hat{j}\right) = \left(-4.25\,\vec{i} - 2.46\,\vec{j}\right) \ m/s^2$$



$$\begin{cases} -R_B \cos 60° - M_3 \ddot{d} + f_d |N_B| - M_3 g \cos 60° = 0 \\ R_B \sin 60° + N_B - M_3 g \sin 60° = 0 \end{cases}$$

$$N_B = \left(M_3 g - R_B\right) \sin 60°$$

$$-R_B \cos 60° - M_3 \ddot{d} + f_d \left(M_3 g - R_B\right) \sin 60° - M_3 g \cos 60° = 0$$

$$R_B = \frac{-M_3 \ddot{d} + f_d M_3 g \sin 60 - M_3 g \cos 60°}{\cos 60° + f_d \sin 60°} = 21.52 \ N$$

$$N_B = 15.35 \ N$$

$$\frac{dE_c}{dt} = J_1 \ddot{\alpha} \dot{\alpha} + M_3 \ddot{d} \dot{d} + J_2 \dot{\omega}_2 \omega_2 + M_2 a_2 v_2$$

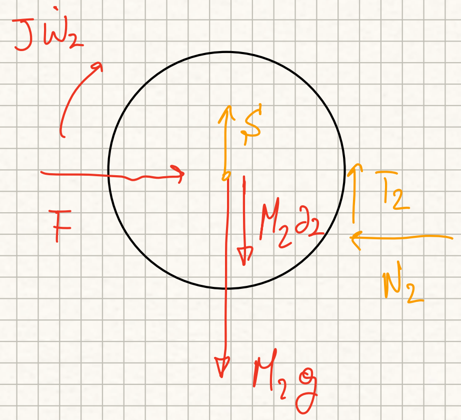
$$v_2 = -\dot{\alpha} R_1 = -1 \ m/s \quad , \quad a_2 = -\ddot{\alpha} R_1 = -1 \ m/s^2$$

$$v_2 = \omega_2 R_2 \Rightarrow \omega_2 = 2 \ rad/s \quad , \quad \dot{\omega}_2 = 2 \ rad/s^2$$

$$\Rightarrow \frac{dE_c}{dt} = 41.11 \ W$$

$$\sum W_{ATT} = C_m \dot{\alpha} \underbrace{\overbrace{-f_d N_B |v_B|}^{-20.74} \overbrace{- f_v R_2 F |\omega_2|}^{-1.5} + \overbrace{M_2 g |v_2|}^{19.62} + \overbrace{M_3 g |v_{By}|}^{37.87}}_{35.25}$$

$$C_m \alpha + 35.25 = 41.11 \Rightarrow C_m = 5.85 \ Nm$$



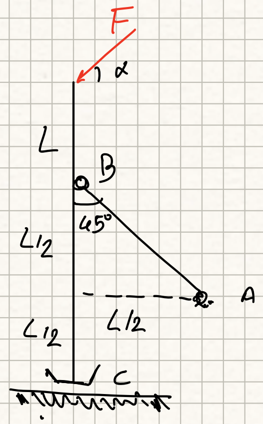
$$N_2 = F = 30 \ N$$

$$T_2 R_2 - J_2 \dot{\omega}_2 - f_v R_2 F = 0 \Rightarrow T_2 = 2.3 \ N$$

$$S = M_2 g - T_2 + M_2 a_2 = 15.32 \ N$$

$$\left(S = M_2 g - f_v F - \frac{J_2}{R_2} \ddot{\alpha} R_1 - M_2 \ddot{\alpha} R_1 \right)$$

ES 2

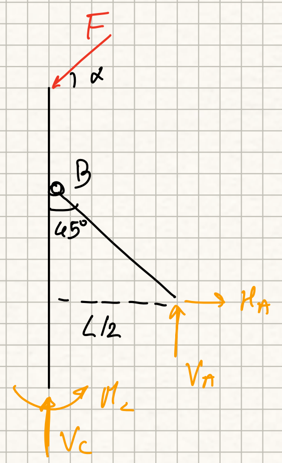


gdl = 3n = 6

gdv = $2_A + 2_B + 2_C = 6$

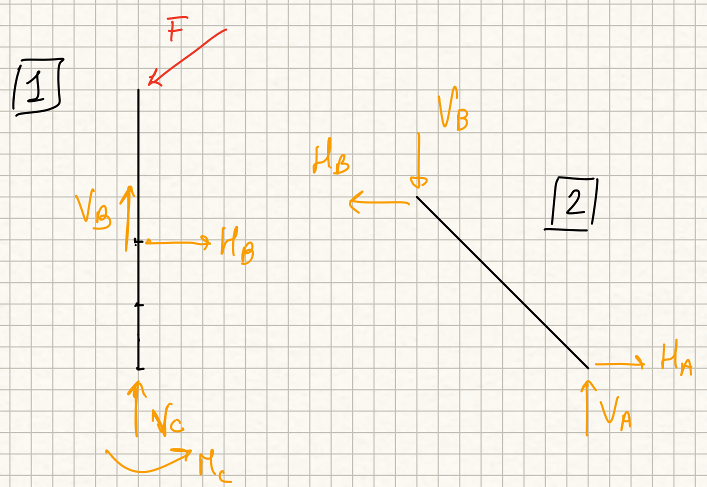
ISOSTATICA

L'asta AB è scarica, ma faccio finta di niente



$H_A - F\cos\alpha = 0 \Rightarrow \boxed{H_A = F\cos\alpha}$

Le altre 2 eqn non servono



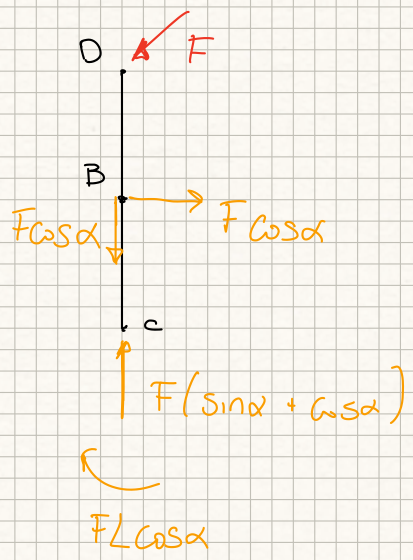
$$\boxed{1}\ \begin{cases} \boxed{H_B = F\cos\alpha} \\ V_c + V_B - F\sin\alpha = 0 \\ \Sigma M_c = 0 \Rightarrow M_c - H_B L + F\cos\alpha\, 2L = 0 \end{cases}$$

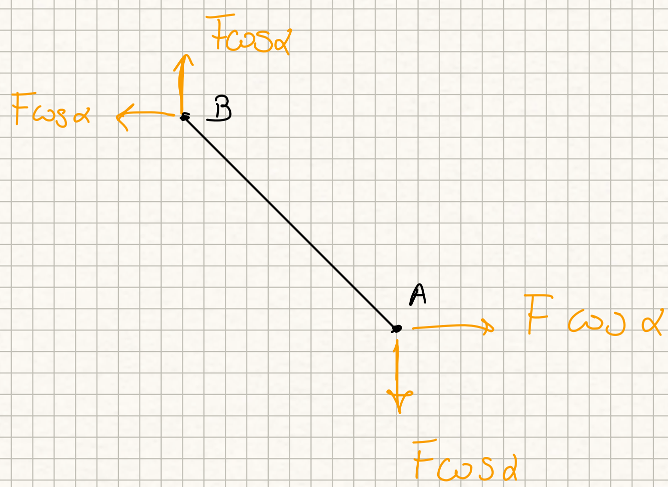
$$M_c - F\cos\alpha\, L + F\cos\alpha\, 2L = 0$$

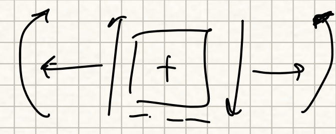
$$\boxed{M_c = -FL\cos\alpha}$$

$$\boxed{2}\ \begin{cases} H_A - H_B = 0 \Rightarrow \boxed{H_A = F\cos\alpha} \\ V_A - V_B = 0 \\ \Sigma M_A = 0 \Rightarrow V_B \frac{L}{2} + H_B \frac{L}{2} = 0 \Rightarrow \boxed{V_B = -H_B = -F\cos\alpha} \end{cases}$$

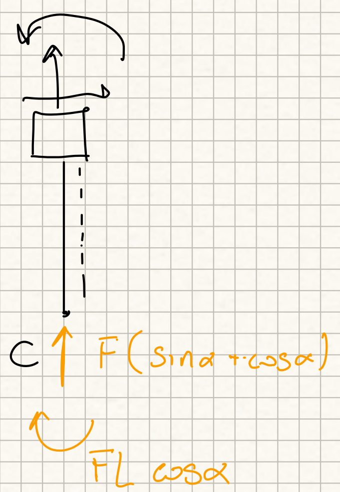
$(2.2) \Rightarrow \boxed{V_A = -F\cos\alpha}$, $(1.2) \Rightarrow \boxed{V_c = F\sin\alpha - V_B = F(\sin\alpha + \cos\alpha)}$





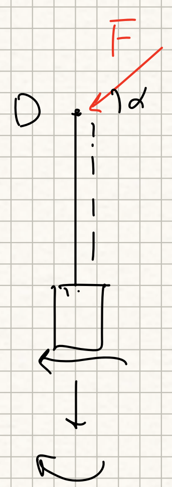


$\boxed{CB}$



$$\begin{cases} N = -F(\sin\alpha + \cos\alpha) \\ T = 0 \\ M_f = FL\cos\alpha \end{cases}$$

$\boxed{DB}$

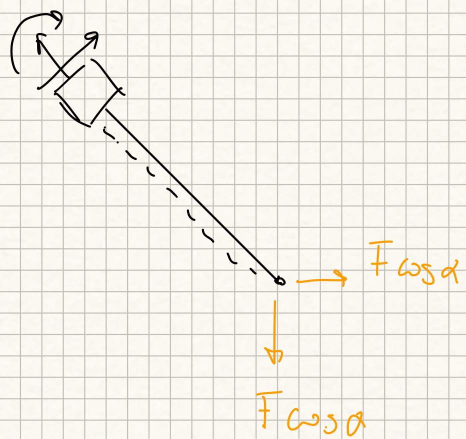


$$\begin{cases} N = -F\sin\alpha \\ T = -F\cos\alpha \\ M_f = F\cos\alpha \, x \end{cases}$$

$M_f(0) = 0$

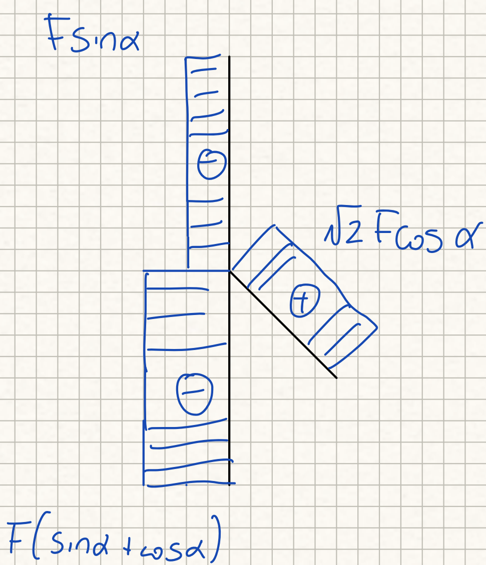
$M_f(L) = FL\cos\alpha$

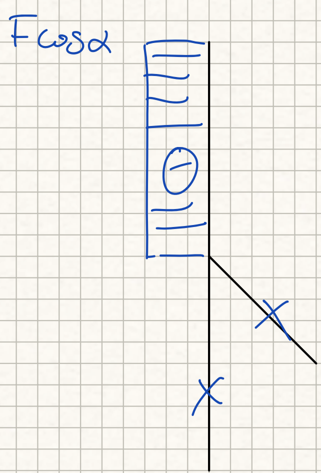
AB

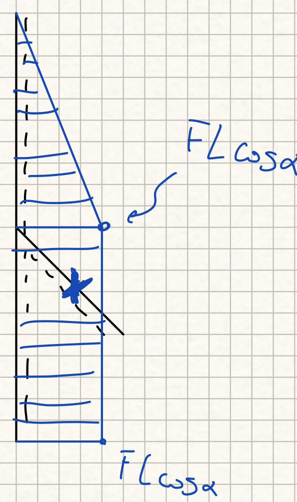


$$\begin{cases} N = \sqrt{2}\, F \cos\alpha \\ T = 0 \\ M = 0 \end{cases}$$

(giusto poiché scarico)

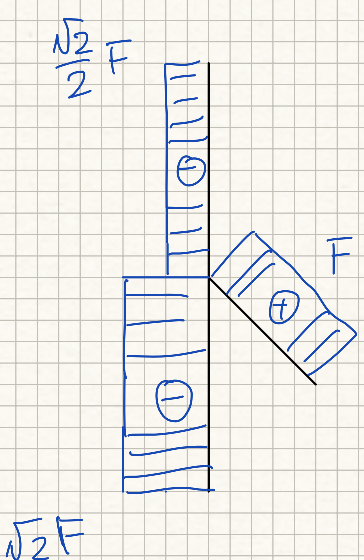


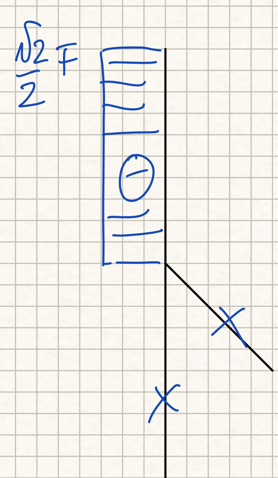


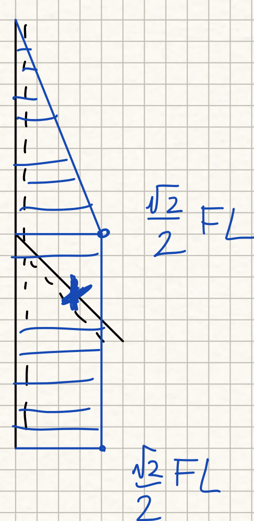


Diciamo $\alpha = 45°$: $H_A = \frac{\sqrt{2}}{2} F$, $V_A = -\frac{\sqrt{2}}{2} F$, $H_B = \frac{\sqrt{2}}{2} F$, $V_B = -\frac{\sqrt{2}}{2} F$

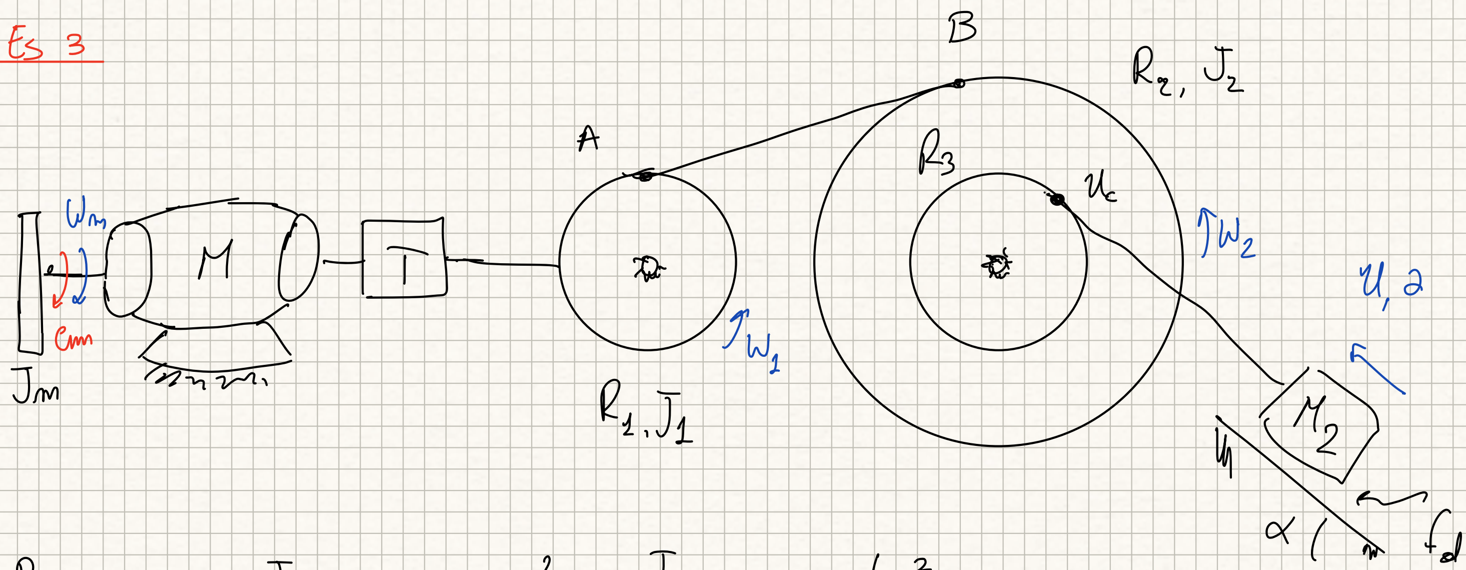
$V_C = F\sqrt{2}$ $\quad M_C = -\frac{\sqrt{2}}{2} FL$







Es 3



$R_1 = 0.5$ m, $J_1 = 1.5$ kg m^2, $J_m = 0.5$ kg m^2

$R_2 = 3$ m, $J_2 = 4$ kg m^2, $R_3 = 1.5$ m, $M_2 = 150$ kg, $\alpha = 10^\circ$, $f_d = 0.7$

$C_{m0} = 60$ Nm, $\omega_0 = 250$ rad/s, $\tau = 0.05$, $\eta_D = 0.9$, $\eta_R = 0.8$

Legami cinematici

$$\omega_1 = \tau \omega_m \Rightarrow \dot{\omega}_1 = \tau \dot{\omega}_m$$

$$v_A = \tau \omega_m R_1, \quad v_B = \omega_2 R_2 \Rightarrow \omega_2 = \tau \omega_m \frac{R_1}{R_2}, \ \dot{\omega}_2 = \tau \dot{\omega}_m \frac{R_1}{R_2}$$

$$v_c = \tau \omega_m \frac{R_1}{R_2} R_3 \Rightarrow v_M = \tau \omega_m \frac{R_1}{R_2} R_3, \quad a_m = \tau \dot{\omega}_m \frac{R_1}{R_2} R_3$$

LATO MOTORE (W_1)

$$W_1 = C_m \omega_m - J_m \dot{\omega}_m \omega_m$$

LATO UTILIZZATORE (W_2)

$$\frac{dE_c^{(U)}}{dt} = J_1 \dot{\omega}_1 \omega_1 + J_2 \dot{\omega}_2 \omega_2 + M_2 v a = \left[J_1 \tau^2 + J_2 \tau^2 \frac{R_1^2}{R_2^2} + M_2 \tau^2 \frac{R_1^2}{R_2^2} R_3^2 \right] \dot{\omega}_m \omega_m$$

$$\frac{dE_c^{(U)}}{dt} = J_u^* \dot{\omega}_m \omega_m \quad \text{dove} \quad J_u^* = 0.0275 \text{ kg m}^2$$

$$\Sigma W_{ATT}^{(U)} = -f_d M_2 g \cos\alpha \, v_M - M_2 g \sin\alpha \, v_m = -M_2 g \left[f_d \cos\alpha + \sin\alpha \right] \tau \frac{R_1}{R_2} R_3 \omega_m =$$

$$\Sigma W_{ATT}^{(U)} = C_u^* \omega_m \quad \text{dove} \quad C_u^* = -15.87 \text{ Nm} \Rightarrow W_2 = C_u^* \omega_m - J_u^* \dot{\omega}_m \omega_m$$

CASO A : a_M allo spunto

$$W_2 = \underbrace{C_u^* \omega_m}_{<0} - \underbrace{J_u^*}_{>0} \underbrace{\dot{\omega}_m}_{>0} \omega_m < 0$$

>0 poiché in salita

DIRETTO : $\eta_D W_1 + W_2 = 0 \Rightarrow \eta_D \left[C_{m0} - J_m \dot{\omega}_m \right] \omega_m + \left[C_u^* - J_u^* \dot{\omega}_m \right] \omega_m = 0$

$$\eta_D C_{m0} - \eta_D J_m \dot{\omega}_m + C_u^* - J_u^* \dot{\omega}_m = 0 \Rightarrow \dot{\omega}_m = \frac{\eta_D C_{m0} + C_u^*}{\eta_D J_m + J_u^*} = 79.85 \ rad/s^2$$

$$a_M = \tau \dot{\omega}_m \frac{R_1}{R_2} R_3 = 0.998 \ m/s^2$$

CASO 2 : C_m, ω_m a regime

$$W_2 = C_u^* \omega_m < 0 \Rightarrow \text{DIRETTO}$$

$$\eta_D C_m \omega_m + C_u^* \omega_m = 0 \Rightarrow C_m = 17.63 \ Nm$$

$$C_m - C_{m0} = -\frac{C_{m0}}{\omega_s} (\omega_m - 0) \Rightarrow \omega_m = \frac{(C_{m0} - C_m)\omega_s}{C_{m0}} = 176.5 \ rad/s$$

CASO 3 : C_m per accelerazione imposta $a_M = 0.5 \ m/s^2$

$$\dot{\omega}_m = \frac{R_2 a_M}{\tau R_1 R_3} = 40 \ rad/s^2$$

$$W_2 = C_u^* \omega_m - J_u^* \dot{\omega}_m \omega_m > 0 \quad \text{DIRETTO}$$

$$\eta_D \left[C_m - J_m \dot{\omega}_m \right] \omega_m + \left[C_u^* - J_u^* \dot{\omega}_m \right] \omega_m = 0$$

$$C_m = \frac{\eta_D J_m \dot{\omega}_m - C_u^* + J_u^* \dot{\omega}_m}{\eta_D} = 38.86 \ Nm$$